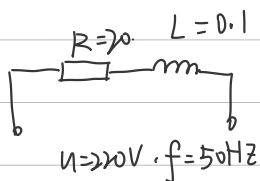


2.3.5



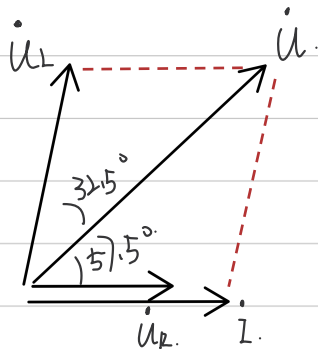
$$X_L = \omega \cdot L = 2\pi f \cdot L = 31.42 \Omega$$

$$Z = \sqrt{X_L^2 + R^2} \approx 37.25 \Omega$$

$$I = U/Z = 220/37.25 \text{ A} \approx 5.91 \text{ A}$$

$$U_R = I \cdot R \approx 118.2 \text{ V}$$

$$U_L = I \cdot X_L \approx 185.6 \text{ V}$$



发现还是最好用相量法做，这样能看出角度：

$$\text{设 } \dot{U} = 220 \angle 0^\circ$$

$$\dot{I} = \dot{U} / (R + jX_L) = 5.91 \angle -57.5^\circ \text{ A}$$

$$\dot{U}_R = \dot{I} \cdot R = 118.2 \angle -57.5^\circ \text{ V}$$

$$\dot{U}_L = \dot{I} \cdot X_L = 185.6 \angle 32.5^\circ \text{ V}$$

2.3.6

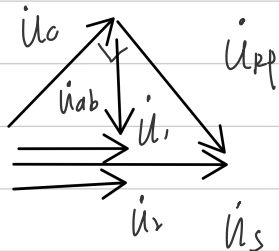
$$\text{线圈电阻 } R = U/I_1 = 120/20 \cdot \Omega = 6 \Omega$$

$$\text{线圈阻抗模 } |Z| = U/I_2 = 220/28.2 \Omega = 7.8 \Omega$$

$$\text{线圈感抗 } X_L = \sqrt{Z^2 - R^2} = 4.984 \Omega$$

$$\text{线圈电感 } L = X_L / \omega = X_L / (2\pi f) = 15.9 \text{ mH}$$

2.3.8 (1)



$$(2) \begin{cases} \dot{U}_1 + \dot{U}_2 = \dot{U}_s = 2\dot{U}_1 = 2\dot{U}_2 \\ \dot{U}_s = \dot{U}_{RP} + \dot{U}_0 \Rightarrow \dot{U}_{ab} = \frac{1}{2}\dot{U}_s - \frac{1}{1+j\omega R_p} \dot{U}_s \\ -\dot{U}_0 - \dot{U}_{ab} + \dot{U}_1 = 0 \end{cases}$$

$$= \frac{j-1}{2(1+j)} \dot{U}_s$$

$$= \frac{1}{2} j \dot{U}_s$$

即. U_{ab} 有效值是 U_s 的一半.
相位超前 U_s 90° .

$$1.3 \Rightarrow \dot{U}_{ab} = \frac{U_s}{2} \angle 2\rho$$

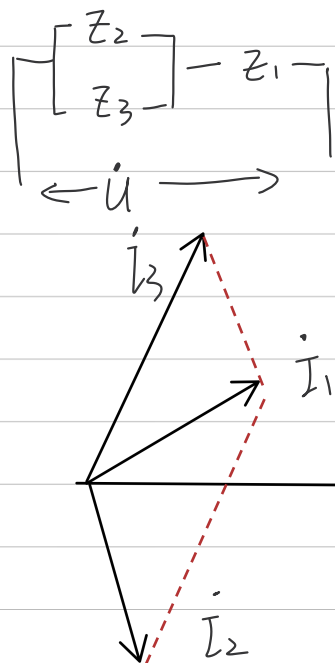
$$\varphi = \arctan \frac{X_C}{R_p}$$



当 $R_p \rightarrow \infty$ 时, U_{ab} 有效值不变
 U_{ab} 与 U_s 相位差从 $180^\circ \rightarrow 0^\circ$

2.3.9.

(1)



$$\dot{I}_1 = \dot{U} \left(Z_1 + \frac{Z_2 Z_3}{Z_2 + Z_3} \right) = 44 \angle 36.8^\circ \text{ A}$$

$$\dot{I}_2 = \dot{I}_1 \frac{Z_3}{Z_2 + Z_3} = 53 \angle -53.13^\circ \text{ A}$$

$$\dot{I}_3 = \dot{I}_1 \frac{Z_2}{Z_2 + Z_3} = 55 \angle 73.74^\circ \text{ A}$$

(2)

有功功率: $P = UI_1 \cos \varphi = 220 \times 44 \times \cos 36.8^\circ = 7744 \text{ W}$

无功功率: $Q = UI_1 \sin \varphi = 5808 \text{ Var}$

视在功率: $S = UI_1 = 9680 \text{ V} \cdot \text{A}$

功率因数: $\lambda = \cos \varphi = 0.8$

2.3.11.

(1) $\lambda = \frac{W}{U \cdot I} = \frac{40}{220 \times 0.41} = 0.4435$

(2) $X_C = \frac{1}{\omega C} \quad I_C = j\omega C U$

$$\dot{I}_C + \dot{I} = \dot{I}' \quad \dot{I} = I (\cos \varphi - j \sin \varphi)$$

$$= 2\lambda - \sqrt{1-\lambda^2} j$$

$$\therefore \dot{I}' = 0.186 \angle 12.2^\circ \text{ A}$$

$$\lambda' = \cos(-12.2^\circ) = 0.977$$

